

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2013

CL 425 Soil Erosion and Conservation

Date: 11.05.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Enlist and discuss in detail factors affecting soil erosion. [07]
- Q - 2(a) 1. What are the effects of soil erosion? [02]
2. What are the biological methods of water erosion control? [02]
- (b) Briefly discuss the principles of water erosion control. [04]
- (c) Write short notes on cropping system and strip cropping. [06]

OR

- Q - 2(a) 1. Explain process of soil erosion briefly. [02]
2. Give detailed classification of mechanical methods of water erosion control. [02]
- (b) Define waterways. Enlist the data required to design it. Write the design steps. [04]
- (c) What is contour bunding? Which types of surplus arrangements are suitable for removing excess water from it? [06]
- Q - 3 (a) Differentiate temporary and permanent structures provided to conserve the soil and water. [04]
- (b) What precautions and measures should be taken to protect the forest? [04]
- (c) Discuss in detail the role of forests in soil conservation. [06]

OR

- Q - 3 (a) Draw typical profile of gully showing permanent structure. [04]
- (b) How the reclamation of gullies has been carried out? [04]
- (c) How loose boulder check dams are constructed for conservation of soil in hilly areas? [06]
Make a neat sketch of loose boulder check dam.

SECTION – II

- Q - 4 (a) List out the principles of wind erosion control. [02]
- (b) Enlist various factors affecting erosion by wind. [05]
- Q - 5 (a) Explain the mechanics of wind erosion in detail. [07]
- (b) Which control practices have been taken to reduce soil erosion by wind on dry land cultivated soil? Discuss any four in detail. [07]

OR

- Q - 5 (a) Discuss in detail all variables affecting in estimation of soil erosion by wind. [07]
(b) Explain the methods of measures to control soil erosion by wind on irrigated land, vegetable and crop land in detail. [07]
- Q - 6 (a) Enlist various factors affecting erosion by water. Discuss each in detail. [04]
(b) How the soil erosion is affected by raindrop? [04]
(c) Differentiate following with figures: [06]
(i) Rill erosion and Gully erosion (ii) Stream erosion and Water fall erosion

OR

- Q - 6 (a) Write short notes on following: [06]
(i) splash erosion (ii) Sheet erosion
(b) The row crop is grown for 1 year period for terraces with underground outlets. [08]
Determine the soil loss from an area for data given below:
Rainfall & runoff erosivity index = 3000
Soil erodibility factor = 0.02 Mg/ha-year
Field slope = 2% and length of slope = 90 m
Cover management factor = 0.16

Criteria for Example [Q-6(b)]

(1) Slope stiffness factor (S):

- $s < 4$ m long, $S = 3.0 (\sin \theta)^{0.5} + 0.56$
 $s > 4$ m long & $< 9\%$, $S = 10.8 (\sin \theta) + 0.03$
 $s > 4$ m long and $s \geq 9\%$, $S = 16.8 (\sin \theta) - 0.50$

(2) Strip Cropping Factor (P_s):

- $P_s = 1.00$ for contouring only or for alternating strips of corn and small grain
= 0.75 for 4-year rotation with 2 years of row crop
= 0.50 with 1 year of row crop

(3) Terrace Sedimentation Factor (P_t):

- $P_t = 1.0$ for no terraces
= 0.2 for terraces with graded channel sod outlets
= 0.1 for terraces with underground outlets

(4) Contouring Factor P_c , Maximum Slope Length and Maximum Strip Crop Width for Different Crops:

Land Slope (%)	P_c value	Maximum Slope Length* (m)	Maximum Strip-Crop Width (m)
1-2	0.6	120	40
3-5	0.5	90	30
6-8	0.5	60	30
9-12	0.6	36	24
13-16	0.7	24	24
17-20	0.8	18	18
21-25	0.9	15	15

*Factor for farming upslope and downslope is 1.0.

*Maximum slope length for strip cropping can be twice that for contouring only.
